

Taher Madraswala

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Profile Summary

AI/ML-focused engineering graduate with hands-on experience in Python, C++, machine learning, deep learning, and full-stack application development. Built and deployed practical projects involving data preprocessing, feature engineering, model training, evaluation, and explainable AI using tools such as PyTorch, TensorFlow, scikit-learn, FastAPI, and React. Strong in problem-solving, clean coding, testing, and collaborative development, with experience across research, software engineering, and AI-driven product prototyping.

Skills

- **Programming Languages:** Java, C++, Python, JavaScript, HTML, CSS, PHP.
- **Artificial Intelligence and Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Model Training Evaluation, Data Preprocessing, Feature Engineering, Explainable AI.
- **AI Frameworks Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, FAISS, OpenAI API, LlamaIndex
- **Software Engineering:** Data Structures Algorithms, Object-Oriented Programming, REST APIs, FastAPI, Flask, Software Testing, Debugging, Authentication, Modular Design.
- **Databases and Search:** SQL, PostgreSQL, MySQL, Semantic Search, Retrieval Pipelines.
- **Developer Tools:** Git, GitHub, Streamlit, Gradio, Docker.
- **Core Concepts:** Model Validation, Performance Evaluation, Data Pipelines, Version Control, Problem Solving, SDLC.

Work Experience

AI Research Intern – Ruby Hall Clinic Feb 2026 – May 2026

- Built a multimodal decision-support system for early diabetic foot risk screening by combining clinical indicators with thermography images to support preventive triage.
- Developed an end-to-end machine learning pipeline covering data preprocessing, feature engineering, model training, and evaluation using classical ML models and an EfficientNetB0-based CNN.
- Improved interpretability and usability through weighted late fusion, SHAP, Grad-CAM, and a Gradio interface to deliver an explainable and practical prototype.

Software Intern – Elite Infosoft May 2026 – July 2026

- Contributed to the development and testing of an Asset Management System used for inventory and asset tracking.
- Assisted with feature implementation, bug fixing, and system enhancements to improve application functionality and user experience.
- Performed functional testing, validated workflows, and identified issues to support stable and reliable software delivery.
- Collaborated with the development team on frontend and backend tasks based on project requirements and testing feedback.

Education

Vishwakarma Institute of Information Technology, Pune, India

B.Tech. in Electronics and Telecommunication May 2026

Samulbai Mulchand Choksey High School and Junior College, Pune, India

HSC XII June 2022

The Bishop’s Co-ed School, Undri, Pune, India

ICSE X July 2020

Project Work

LLM Powered Intelligent Query Retrieval System 📄:

- Built an LLM-powered document retrieval system that converts natural language queries into contextual answers from unstructured documents.
- Developed a React frontend and FastAPI backend with OpenAI, LlamaIndex, and FAISS-based semantic search for fast retrieval over uploaded documents.
- Implemented secure document upload, authentication, and multi-format file handling to support a modular and scalable user workflow.

Omnix: AI Software Delivery Engine 📄:

- Developed an AI agent orchestration framework that converts natural-language project requirements into a structured software

- delivery pipeline using specialized agents for clarification, planning, architecture, coding, review, and validation.
- Implemented modular workflow components including state management, agent routing, checkpointing, recovery handling, logging, and summary generation to ensure reliable and traceable execution.
- Enabled test-driven development workflow with human-in-the-loop approvals, automated validation, rollback support, and configurable execution for scalable and safe project delivery.

**AI-Powered Collateral Valuation and Resale Liquidity Engine **:

- Designed and developed an AI-driven collateral intelligence system that estimates market value, distress sale value, resale potential index, estimated time to liquidate, confidence score, and risk flags for property-backed lending, combining valuation and liquidity analysis in a single framework.
- Built a structured, explainable decision engine using location intelligence, property characteristics, legal and ownership attributes, rental signals, and market dynamics to generate transparent, range-based outputs for lending and risk assessment.
- Implemented a Python-based Streamlit application with modular engines for valuation, liquidity scoring, explainability, vision analysis, and stress simulation, using circle-rate benchmarks, synthetic listings, and locality-level property data across multiple Indian cities..

Achievements

Publications

**Graph Algorithms for Network Analysis **

International Journal of Educational Research

This paper explores how graph algorithms like Dijkstra’s, Prim’s, and PageRank help solve real-world challenges in networks, from social media to communication systems. It highlights their role in optimizing connectivity, detecting communities, and improving data flow.

Certifications

**IBM Full Stack Software Developer **

Coursera, 2025

**Python for Data Science and Machine Learning Bootcamp **

Udemy, 2024

Leadership & Activities

Entrepreneurship Development Cell, VIIT, Pune

Associate Operations and Supply Management Head

- Led a team of 15 members to handle logistics and resource management for Vishwapreneur 24, an event with over 500 attendees.
- Coordinated vendors and optimized setup processes to deliver execution under tight timelines.